

Cloud Storage for Train Data

2026/01/02 12:19:18

Search before you generate an image for a presentation front page that...

Thinking: Thinking: Seeking Visual References I'm currently focused on...



Search before you generate an image for a presentation front page that introduce a cloud storage for Train Data Recorder

Context:

Train Data Recorder are the counterpart of a "blackbox" in a plane, despite the name both are mostly orange boxees with lots of connectors, typical suppliers is Deuta Werk product name Redbox).

The data recording is event driven and monthly the data volume is about 10GB per train.

A train fleet typical consists of up 500 cars, but mostly divided into several classes or lines for commuter trains all having nearly the same train data recorder or at least some comparable protocol structures. Typically a remote data transfer should be assumed, while older trains may need manually upload of data.

Application of cloud storage:

The operating company want to organize data on a central cloud storage provide and want a dashboard the shows the progress of train recording and list of events like accidents or repair and maintenance activities needed. Estimate with 500 trains and 100GB per year and train which results in 50TB anually, therefore database structures where you can move time range like a year without migration are essential (Postgres partitioning).

Dashboard should allow drill down of single train data with maps showing the track and all parameters, like a

train simulator.

Design elements of the image:

Visualize separate fleets, while wikipedia has lots of images, we need something more abstract.

Visualize a could with datatabse.

Visualize trains, side/front view mix, commuter trains and high speed train look very different from side view and front view is typically recognized better.

Dashboard showing maps and speed over time or distance should be visualize. These plots show lots of signal details, examples can be supplied.



Thinking:

Thinking:

Seeking Visual References

I'm currently focused on gathering visual references. I'm actively searching for images and information on train data recorders, specifically the Deuta Werke Redbox. I'm also looking for examples of train fleet management dashboards, cloud storage architectures, and data visualizations related to train metrics like speed and location, to inform the next design step.

Analyzing Visual Data

I've been gathering visual data on train data recorders, particularly the Deuta Werke Redbox, to understand their visual characteristics. In addition, I'm researching train fleet management dashboards, cloud architectures, and data visualization examples that relate to train metrics like speed and

location. This visual exploration will directly inform the forthcoming design phase.

Refining the Concept

I'm currently focused on the core concept: cloud storage for TDR data. I'm breaking down the subject and considering the context of TDRs and cloud solutions to form a strong abstract visual representation. I'm exploring the key elements that need to be conveyed to experts.

Conceptualizing the Design

I'm now integrating the elements from my deconstruction. The abstract fleet visualization is taking shape, aiming to represent different train types without resorting to literal photos. I'm focusing on connecting the cloud icon and database symbol clearly to the TDR "black box," establishing a direct visual link.

Formulating the Visuals

I've been working on the design elements. I'm focusing on the abstract fleet representation, aiming for visual distinction between train types. I'm ensuring a clear visual link between the cloud, database, and the TDR "black box," emphasizing the data flow from trains to the cloud. The dashboard visualization is also taking shape.

Defining the Elements

I'm now integrating the design elements. I'm focusing on the abstract train representations, distinguishing between types visually. I'm working to visually link the cloud and database clearly with the TDR.

Evaluating the Visualization

I'm currently assessing the generated image's alignment with the user's specific request. The visualization seamlessly combines abstract cloud and database elements with a labeled orange Train Data Recorder. I'm carefully

examining the integration of commuter and high-speed train fleets within the scene, and also carefully checking the dashboard overlay.

Assessing the Integration

The generated image now meets the user's detailed specifications. I've confirmed that the abstract cloud and database elements are present as requested, in tandem with the labeled orange Train Data Recorder. I'm satisfied that the commuter and high-speed train fleets, along with the dashboard, are integrated as described. I consider it a good match.



[1]: DEUTA REDBOX® 2K

[2]: DEUTA REDBOX® pro

[3]: Design storage for AI and ML workloads in Google Cloud | Cloud Architecture Center